INCH-POUND

MIL-A-53089
INT. AMENDMENT 1
1 March 1990

MILITARY SPECIFICATION

AIR CONDITIONERS: VERTICAL AND HORIZONTAL,

COMPACT, MULTIPLE POWER INPUT

This amendment forms a part of MIL-A-53089, dated 9 February 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, line 9, under type II, add:

"C - 36,000 Btu/hr - 208V, 3-Phase, 50/60/400 Hz."

Delete lines 10 through 15 in their entirety.

PAGE 2

2.1.1, line 7, add:

"SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
NN-P-71	- Pallet, Material Handing, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals."

PAGE 3

2.1.2, line 14, ADD:

"TA13229E6700	- Air Conditioner, Horizontal, Compact, 36,000 Btu/hr, 208 Volt-AC, 3-Phase, 50/60/400 Hertz."

PAGE 5

3.1, line 7, after "Type II - Horizontal, Size - B", add:

"Size C - 36,000 Btu/hr - TA13229E6700"

AMSC N/A FSC 4120

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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PAGE 6

3.4, line 3, add: "All rubber components shall be ozone resistant and no more than four calender quarters old from the date of manufacture to the date of acceptance by the Government."

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3.5.1.2.2, delete in its entirety and substitute:

"3.5.1.2.2 Type II, sizes B and C. After being subjected to 70 ±5 °F on the front and 0 ±2 °F for size B, or -25 +0, -2 °F for size C, on the rear for a period of 10 hours, the air conditioners shall start and operate in the cooling mode as specified with air at 70 ±5 °F entering the evaporator and air at 0 ±2 °F for size B, or -25 +0, -2 °F for size C, entering the condenser."

PAGE 8

3.5.2, table I, delete in its entirety and substitute:

"TABLE I. Cooling capacities.

Unit	Condition	Ambient temperature (dry bulb)	Evaporator inlet temperature		Minimum net sensible cooling capacity	Minimum net total cooling capacity
			dry bulb	wet bulb		
		°F	°F	°F	Btu/hr	Btu/hr
Type I						
Size C	A	120	90	75	12,100	18,500
	B	95	80	67	14,000	21,500
Type II						
Size B	A	120	90	75	14,000	18,500
	B	95	80	67	14,700	19,500
Size C	A	120	90	75	30,000	41,000
	B	95	80	67	32,000	40,000

Note: Data shown for 0-inch water gage (WG) (free air) external static pressure on the evaporator for type I units and 0.25 inch WG for type II units."

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3.5.3, table II, delete in its entirety and substitute:

"TABLE II. Condensing and suction pressures (psig).

Unit	Operating conditions (see 3.5.1.1)			Operating conditions (see 3.5.2)	
	Condensing pressure (max)	(min)	Suction pressure (min)	Condensing pressure (min)	Suction pressure (min)
Type I					
Size C	470	-	50	440	65
Type II					
Size B	470	160	50	425	50
Size C	470	160	50	430	50

"

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3.5.6, table III, delete in its entirety and substitute:

"TABLE III. Airflow, heating capacity and power consumption.

Unit	Evaporator Airflow (minimum SCFM) (1)	Heating (minimum Btu/hr) (2)	Power Consumption (maximum KW) (3)	Power Factor (minimum) (4)
Type I				
Size C	570	12,000	6.5Kw	0.9
Type II				
Size B	580	13,400	6.5Kw	0.9
Size C	1380	31,200	13.5Kw	0.9

"

3.5.7, line 8, after "Type II, Size B", add:

"Size C - 208V, 3-phase, 50/60/400 Hz.

Total RMS current to ground for type II, size C unit shall not exceed 10mA at full operation when operating at rated voltage."

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3.6.1, table IV, delete in its entirety and substitute:

"TABLE IV. Corrected sound pressure levels in decibels.

Octave band center Frequencies - CPS	Evaporator side (db max)	Condenser side (db max)
Type I		
	Size C	
63	65	60
125	76	73
250	75	74
500	77	71
1000	69	71
2000	66	70
4000	62	69
8000	56	69
Type II		
	Size B	
63	60	60
125	66	64
250	69	63
500	68	65
1000	65	66
2000	60	66
4000	54	64
8000	65	55
	Size C	
63	71	69
125	70	68
250	70	72
500	69	73
1000	76	70
2000	64	65
4000	63	63
8000	53	51

"

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3.9.4, table V, delete in its entirety and substitute:

"TABLE V. High and low cutout pressures.

Unit	Compressor high-pressure cutout (±10 psig)	Compressor low-pressure cutout (±5 psig)
Type I		
Size C	480	7
Type II		
Size B	480	7
Size C	480	15

"

3.10.2, line 2, delete: "of the air condition".

3.15, table VII, delete in its entirety and substitute:

"TABLE VII. Weight.

Type	Size	Maximum Weight (lb)
I	C	280
II	B	290
	C	435

"

4.1.2, delete in its entirety.

4.5.2.1, table VIII, after test req 26, add:

"27	- X -	Electrical power supply."	4.5.2.27	3.5.7

4.5.2.12.1.1, line 1, after "Type II, size B.", add: "and C."

4.5.2.12.1.2, delete in its entirety and substitute:

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"4.5.2.12.1.2 Type II, sizes B and C. Place the air conditioner in a test chamber, installed for in-the-wall operation, with the air conditioner front exposed to a temperature of 70 ±5 °F and the rear (external portion) exposed to 0 ±2 °F for size B, or -25 +0, -2 °F for size C. With power connected to the air conditioner, maintain the temperatures for a 10-hour soak period. Then, write the temperature selection control set for coolest temperature, start and operate the air conditioner in the cool mode for a period of not less than 1 hour with the air at 70 ±5 °F entering the evaporator, and the air at 0 ±2 °F for size B, or -25 +0, -2 °F for size C, entering the condenser. One or more of the following shall constitute failure of this test:

a. Inability of the air conditioner to start and operate in the cooling mode.
b. Temperature differential between the average entering air temperature and average leaving evaporator air temperature less than 15 °F when operating in the cooling mode after stabilization of conditions has occurred.

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4.5.2.19.4, table XIII, delete in its entirety and substitute:

"TABLE XIII. Minimum temperature drop and maximum current draw. 1/

Unit	Input Power	Temperature difference °F (minimum)	Current draw Amps (max)
Type I			
C	208V, 3 phase, 60 Hz	20.0	21
Type II			
B	230V, single phase, 60 Hz	20.0	35
	208V, 3 phase, 60 Hz	20.0	21
C	208V, 3 phase, 60 Hz	20.0	45

1/ For all 3-phase air conditioners, amperage reading shall be the average of the three phases."

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After 4.5.2.26, add:

"4.5.2.27 Electrical power supply. To determine RMS current to ground for the unit, place the air conditioner on a non-conductive surface. Turn power on and measure voltage of each phase with multimeter/voltmeter and record data. Next, attach one of the test lead from digital multimeter (Fluke Model 8050A or equal) to the ground stud of the unit and the other lead to the earth (protective)

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ground. Set the digital multimeter to indicate RMS current rate of 20mA. Turn the unit "ON" to cool position and operate the unit for 15 minutes. The ambient temperature during operation of the unit shall be between 80 and 95 °F. Monitor the digital multimeter to ensure that RMS current is within specified limits. Record data at 5 minutes interval. Shut the unit off and wait for 5 minutes. Repeat this test one more time. Nonconformance to 3.5.7 shall constitute failure of the test.

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5.2.1.2, line 1, delete: "and type II, size B." and substitute: "and type II, sizes B and C."

5.3.1.1, line 1, delete: "and type II, size B." and substitute: "and type II, sizes B and C."

5.3.2.1, line 1, delete: "and type II, size B." and substitute: "and type II, sizes B and C."

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6.9, under "Type II - Horizontal", after line 5, add:

"C 1 - 208V, 3-phase, 50/60 Hz C - 208V, 3-phase, 50/60/400 Hz.
2 - 208V, 3-phase, 400 Hz"

PAGE 45

Figure 1, delete lines which read:

"II A 10.25 ±0.5"

and

"II D 15.75 ±0.5"

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